Euclidifying Gaia

Using Machine Learning to Classify Non-Point-Like Objects in Gaia DR3

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Motivation

The Challenge

Gaia DR3 catalogues 1.8 billion sources but has limited morphological classification

Many extended objects (galaxies, nebulae) are misclassified or unclassified as point sources

Gaia's spatial resolution (~0.1") blurs compact galaxies into star-like profiles

Our Approach

Euclid VIS imaging (0.1"/pixel) resolves structure Gaia cannot

Cross-match Euclid × Gaia to obtain truth labels (star vs galaxy)

Train XGBoost on Gaia-only parameters to classify all Gaia sources without needing Euclid

Key Question: Can Gaia-only parameters, trained with Euclid morphological labels, reliably distinguish stars from galaxies across the whole sky?

The Missions — Gaia & Euclid

Gaia

ESA astrometry mission, launched 2013, L2 orbit

Whole-sky survey: 1.8 billion sources

6-hour spin, 63-day precession → complex scanning pattern

Final shutdown: 27 March 2025

DR3 + future DR4 (2026), final catalogue ~2030

Key limitation: ~0.1" resolution, struggles with morphology

Euclid

ESA dark energy mission, launched 1 July 2023, L2 orbit

Deep-field survey: 14,000 deg² over 6 years

Step-and-stare: ~70 min per field (0.5 deg²) → slow coverage

VIS: 0.1"/pixel, IE-band depth 24.5 AB mag

VIS saturation limit: ~17.5 mag → filters OUT bright sources, biased toward faint/galaxies

Only 17 ERO + 4 Q1 fields (~2000 deg² by Mar 2025, ~14% of total)

The Bridge: Euclid sees deeper and resolves morphology; Gaia sees the whole sky. By training on Euclid labels, we can classify Gaia's entire catalogue — potentially finding many more galaxies than Euclid could ever survey alone.

Data: Euclid ERO & Q1 Releases

ERO (Early Release Observations)

17 fields with VIS-band photometry

Morphological classifiers: class_star, spread_model

Cross-matching: spatial position (q3c) with 0.3" radius

~38,000 matched sources (Gaia G ≥ 18.5)

Highly imbalanced: ~95% stars, ~5% galaxies

Q1 (Quarter 1 Release)

4 fields with multi-band VIS + NIR photometry

PHZ pipeline: photo-z & star probability (phz_star_prob)

Pre-computed gaia_id → direct ID matching (no spatial cross-match)

~30,000 matched sources (Gaia G ≥ 18.5)

Better balanced: ~75% stars, ~25% galaxies

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Cross-Matching

Building the training set from Euclid × Gaia

Cross-Matching Methodology

ERO: Spatial Matching

q3c library: match Euclid × Gaia by sky position

Radius: 0.3" (arcsecond) — optimized for Euclid astrometric accuracy

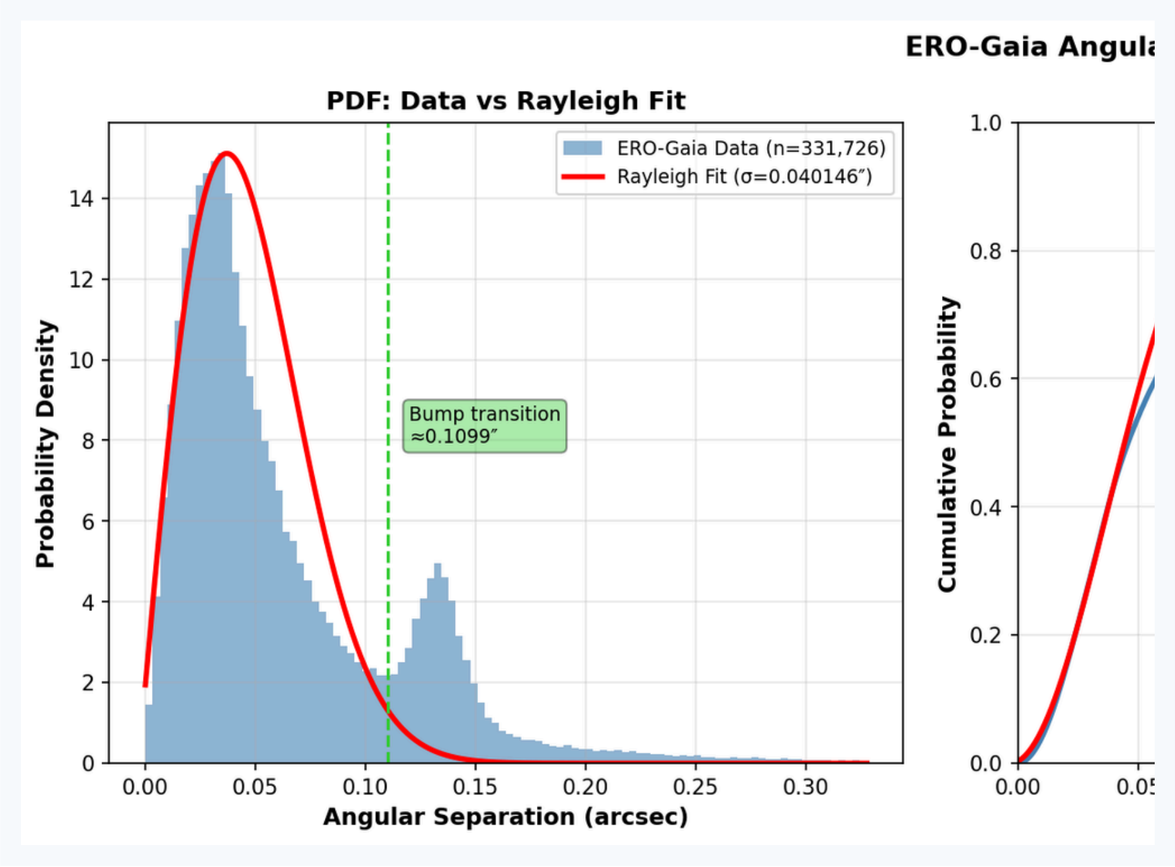
One-to-one pairing: nearest neighbour within radius

Q1: Direct ID Matching

Pre-computed by Euclid team: gaia_id → direct link

More reliable than spatial matching (no radius ambiguity)

Fewer contamination issues



Separation distribution (right): Most matches cluster near 0" with Rayleigh tail. Main peak = true matches; tail = spatially blended/unmatched sources.

Cross-Match Validation: NGC 6397 Discovery

Bimodal separation distribution

Angular separation at 0.3" radius showed an unexpected second peak

Root cause: NGC 6397 globular cluster members have large, coherent proper motions that shifted their epoch-2016 positions

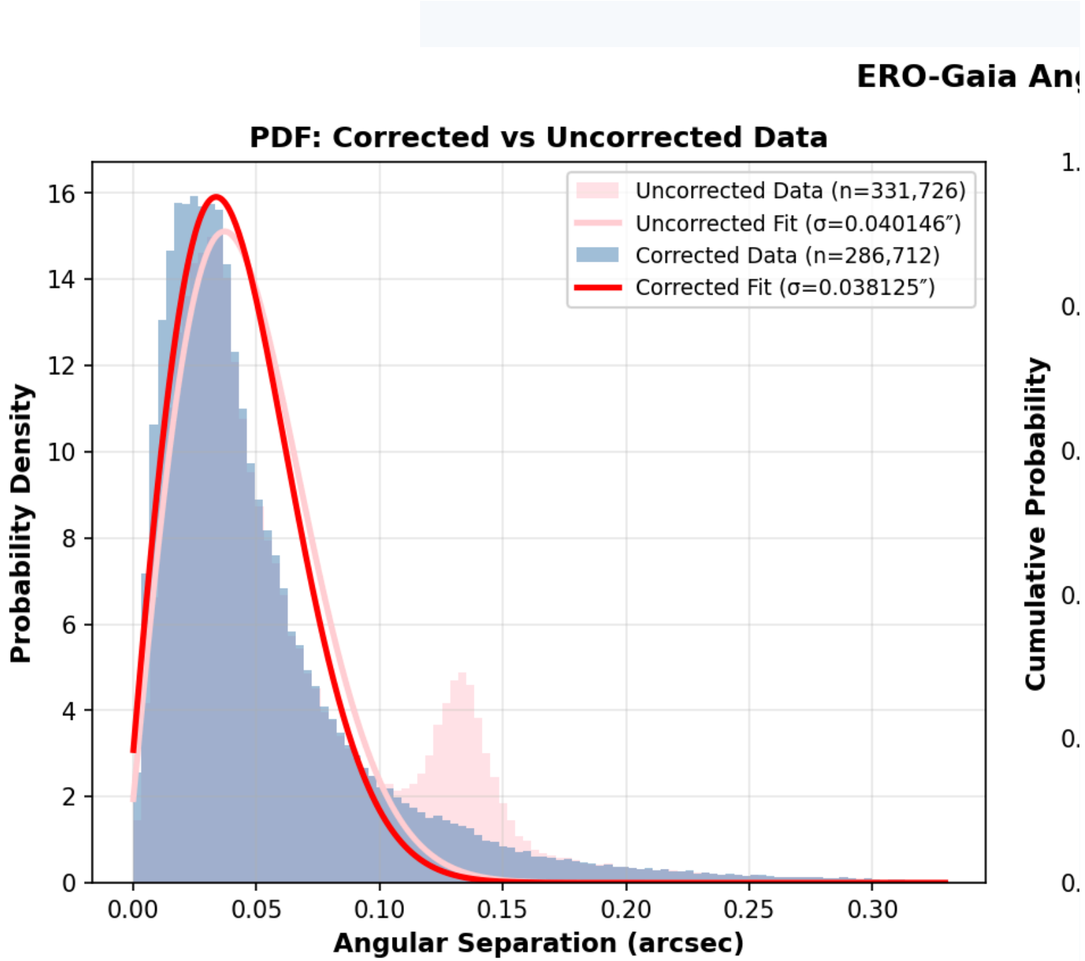
Resolution: Applying proper motion correction collapses the second bump into the main Rayleigh peak

σ = 0.038" (corrected) vs 0.040" (uncorrected)

→ Validates the 0.3" matching radius



NGC 6397



ERO Label Reliability

spread_model

SNR-dependent bias: unreliable at low signal-to-noise

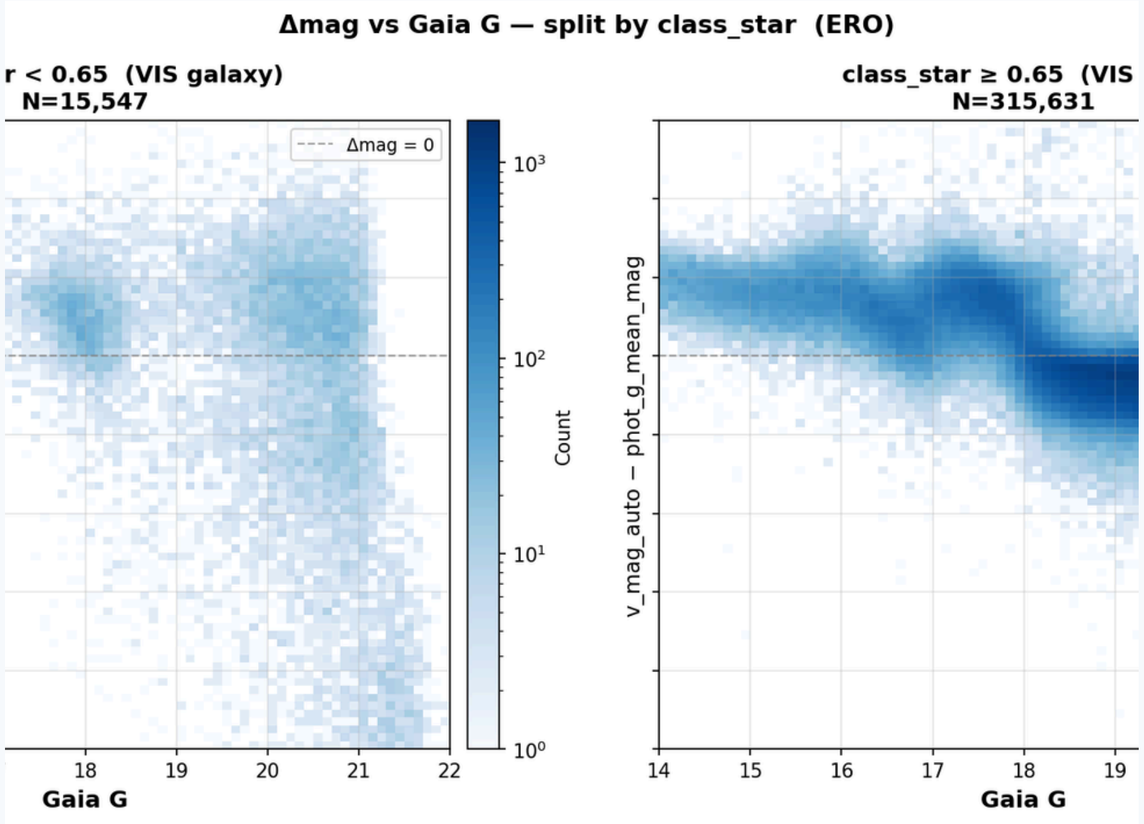
Deserted early in the project

class_star

Suspicious Δmag vs class_star distributions — potential systematics

Under active investigation with supervisors

Implication: ERO labels may have higher contamination than expected, affecting any model trained on them



Ongoing work: Evaluating whether to also desert class_star labels and rely solely on Q1 PHZ-based classifications

Q1 PHZ Validation

PHZ Pipeline

Multi-wavelength photometry → photo-z + classification

phz_star_prob: probability of being a star (0–1)

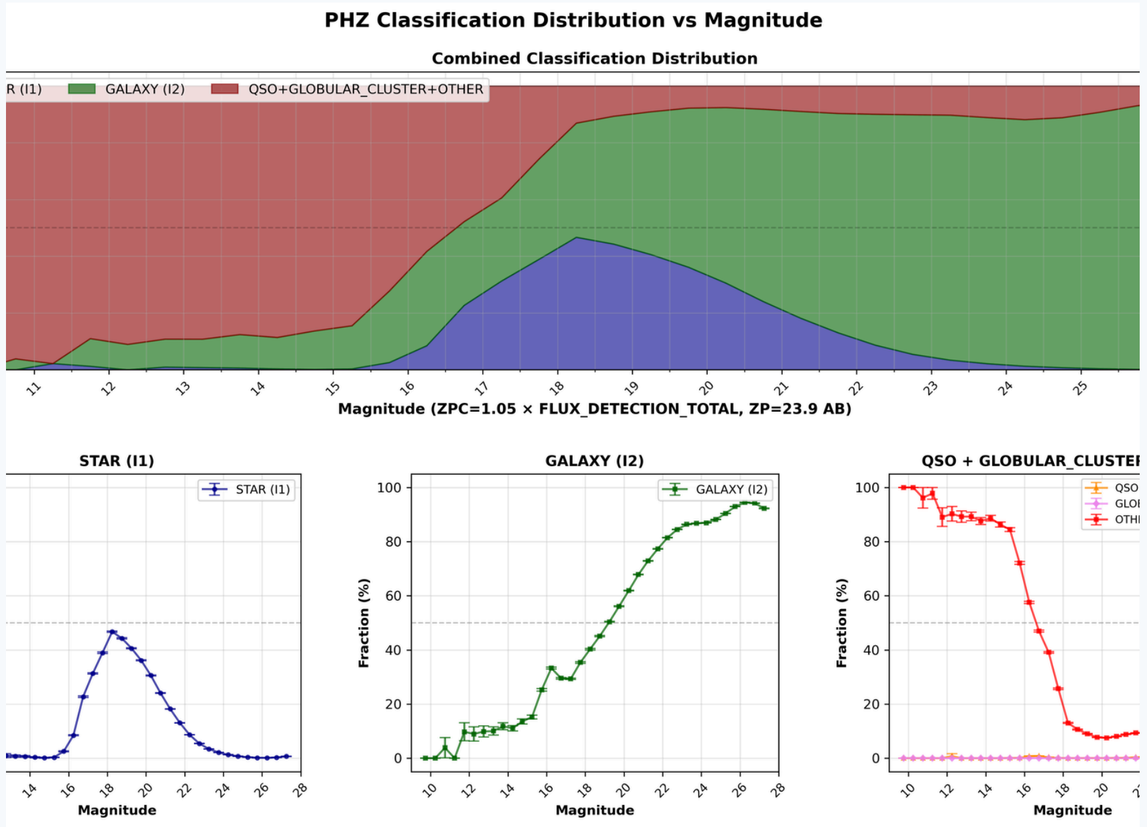
Probabilistic approach reduces hard classification errors

Advantages over ERO classifiers:

More reliable than morphological spread_model

Better class balance (~75% stars, ~25% galaxies)

Physics-informed: trained on SED fitting



Result: Q1 labels are of significantly higher quality than ERO. Models trained on Q1 consistently outperform ERO-trained models.

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XGBoost Classification

Training strategy, feature selection & results

Training Strategy & Features

23 Gaia-only Parameters

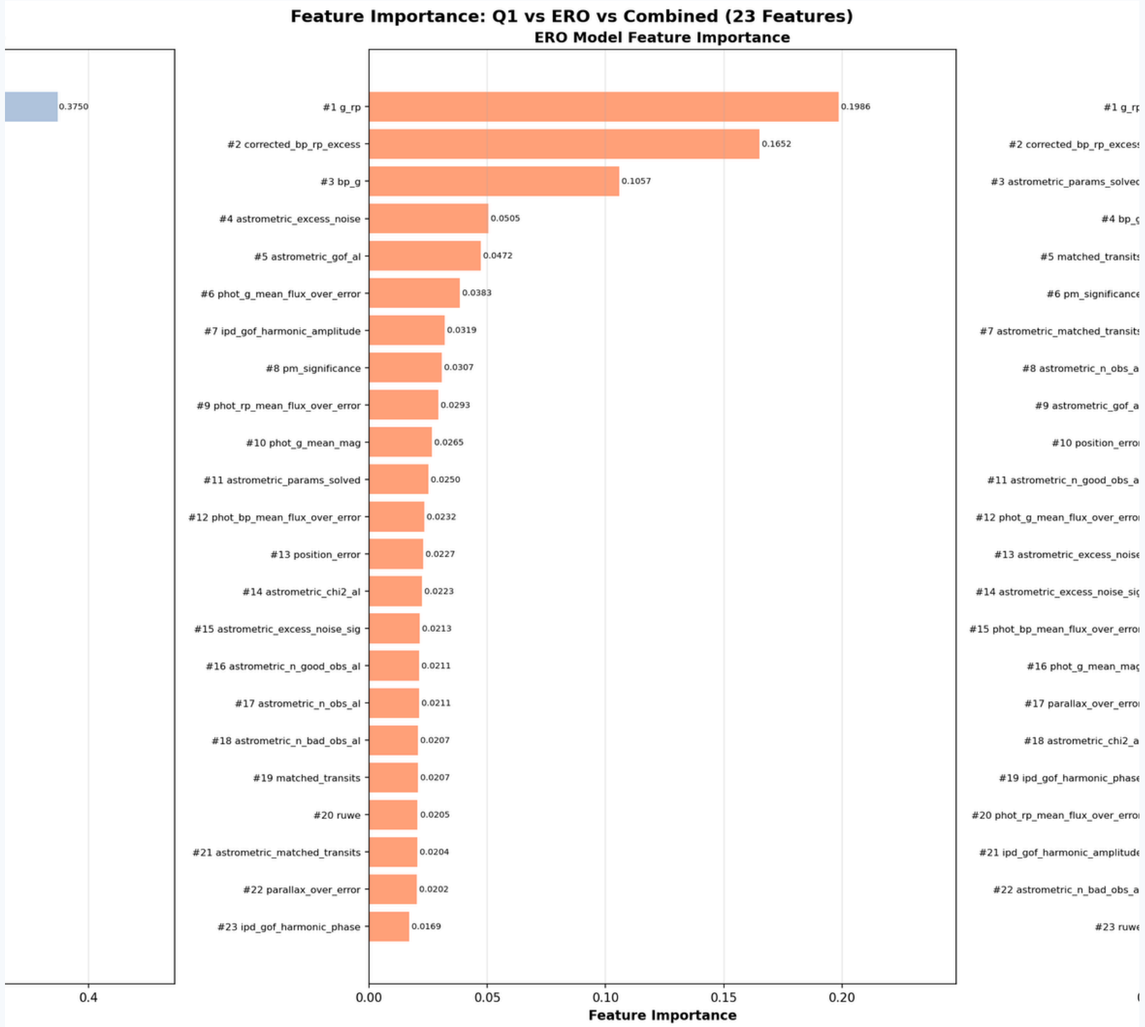
Astrometric: parallax_over_error, ruwe, astrometric_excess_noise, matched_transits, ...

Photometric: phot_g_mean_mag, bp_g, g_rp, corrected_bp_rp_excess, flux SNRs

Kinematic: pm_significance, position_error

Training Strategy

3 training sets: Q1-only, ERO-only, Combined
3 test sets: Q1, ERO, Combined
→ **27 experiments** (9 per training source)
Magnitude cut: Gaia G ≥ 18.5

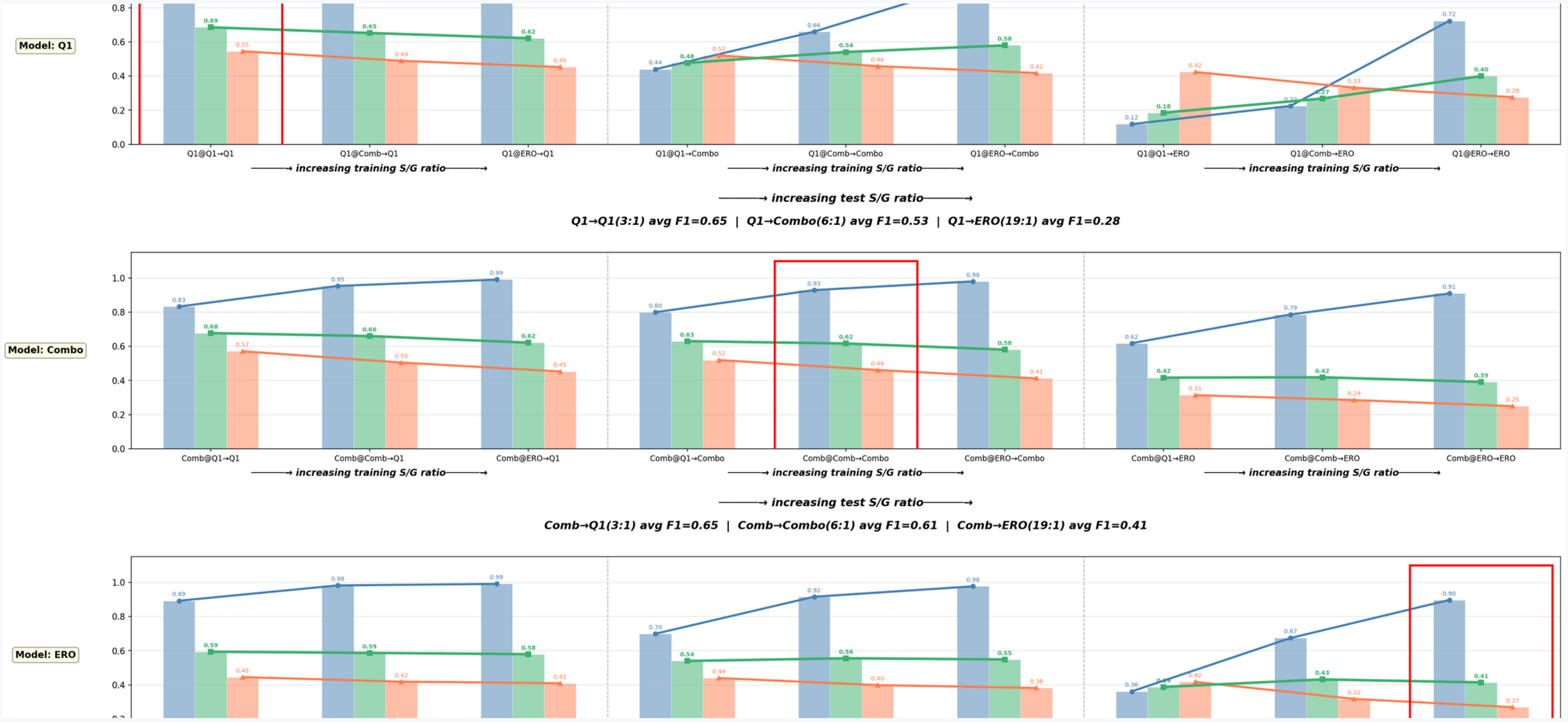


Classification Results

Experiment	Accuracy	Galaxy Precision	Galaxy Recall	Galaxy F1	ROC AUC
Q1 → Q1	0.875	0.922	0.545	**0.685**	**0.876**
Comb → Q1	0.870	0.953	0.505	0.660	0.860
Comb → Combo	0.920	0.929	0.461	0.616	0.867
ERO → ERO	0.962	0.897	0.270	0.415	0.816
Q1 → ERO	0.812	0.118	0.424	0.185	0.678

Key finding: Q1-trained model achieves best ROC AUC (0.876) and galaxy F1 (0.685). Galaxy recall remains the core challenge (~50-55%). Cross-domain transfer (Q1→ERO) degrades significantly, suggesting label-quality differences.

Galaxy Metrics Across Experiments



Feature Importance & Physical Interpretation

Top discriminating features:

1. corrected_bp_rp_excess

Galaxies have extended emission → excess flux in BP/RP relative to G

2. g_rp (colour)

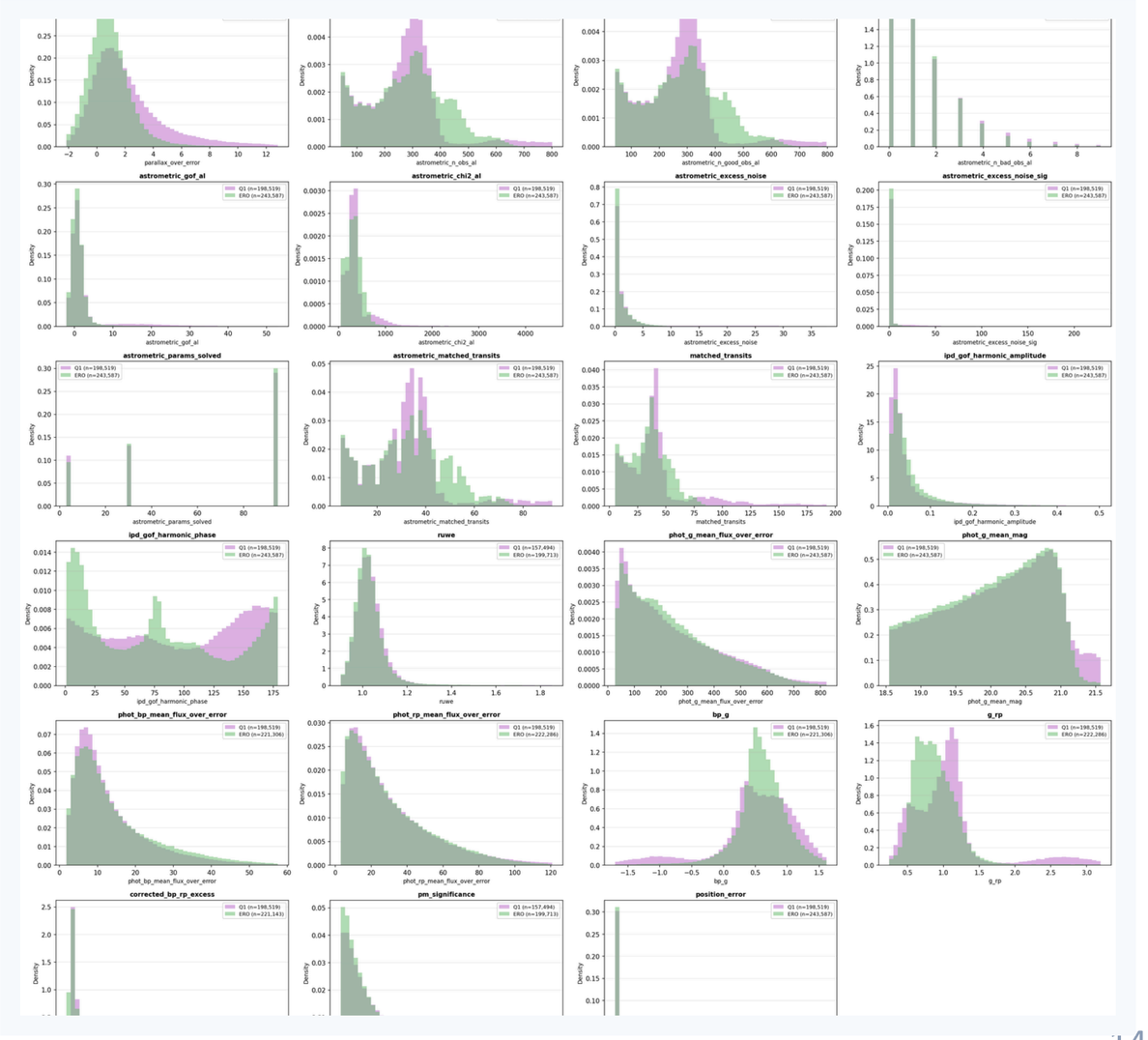
Galaxy spectral energy distributions differ from stellar colours

3. astrometric_matched_transits

Extended sources affect centroiding → fewer matched transits

4. ruwe

Galaxies produce poor astrometric fits → higher RUWE values

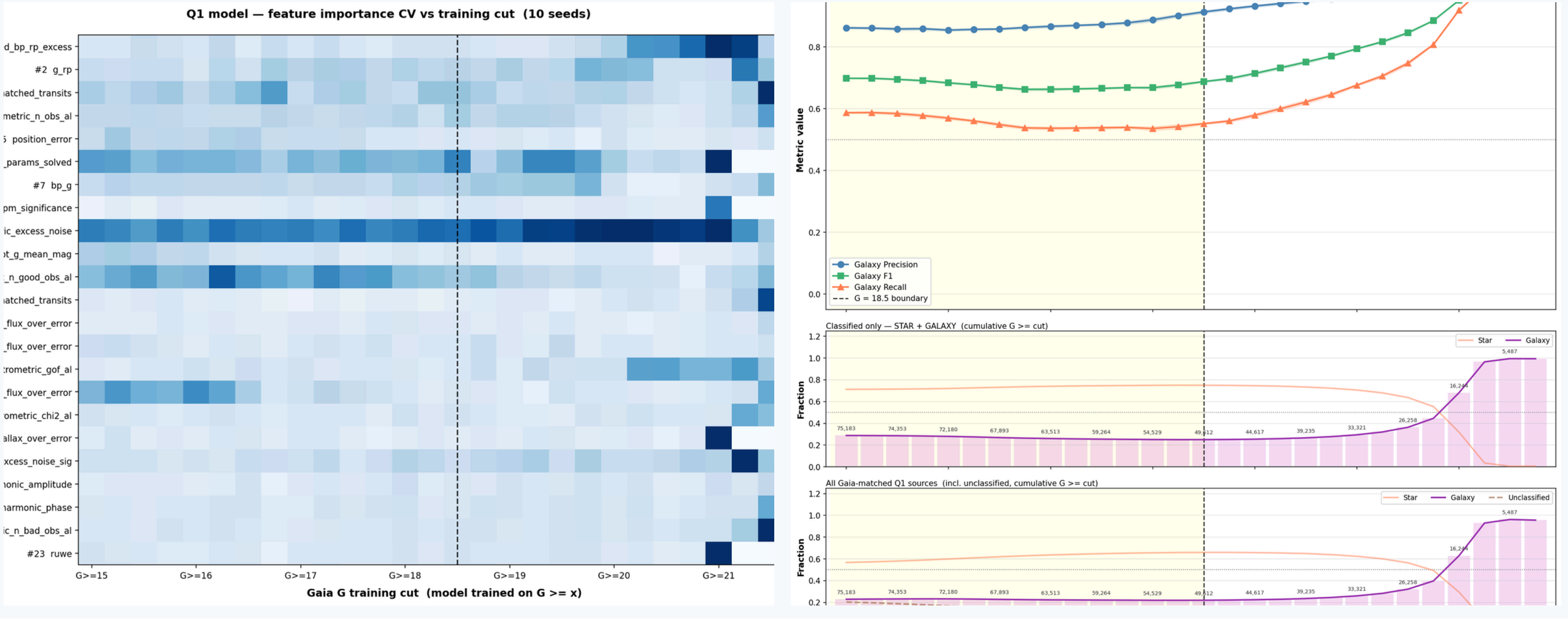


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Sensitivity & Visualization

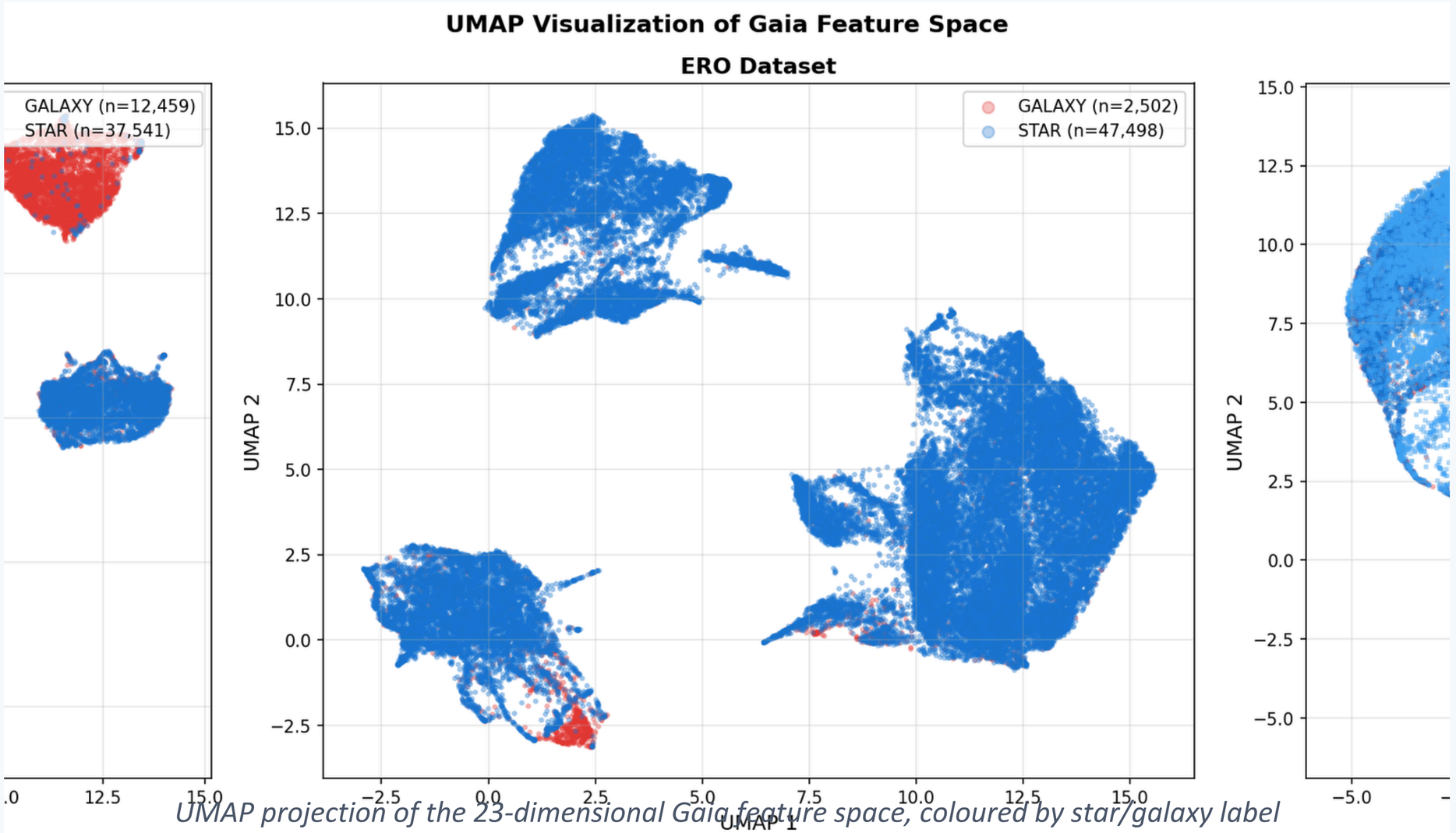
Magnitude cuts and model robustness

Magnitude Cut Sensitivity



Q1 cross-validation performance as a function of magnitude cut (left: heatmap, right: line plots)

UMAP: Star-Galaxy Separation in Feature Space



UMAP projection of the 23-dimensional Gaia feature space, coloured by star/galaxy label

Conclusions & Outlook

Key Results

Euclid × Gaia cross-matching successfully generates morphological training labels

XGBoost achieves ROC AUC 0.88 using 23 Gaia parameters alone

corrected_bp_rp_excess is the strongest predictor (physically motivated)

NGC 6397 proper motion correction validates cross-match methodology

ERO label quality is a concern — Q1 labels are more reliable

Future Directions

Investigate class_star systematics further

Test with future Euclid data releases (larger sky coverage)

Explore galaxy sub-classification (elliptical vs spiral)

Improve galaxy recall through class-weighting or threshold tuning

Apply classifier to full Gaia DR3 catalogue

Bottom line: Gaia parameters contain real morphological signal. With better labels from future Euclid releases, this approach could reliably flag extended objects across the full Gaia sky.

Thank You

Questions?

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